Cryptocurrency Volatility Prediction Using Multimodal Deep Learning: Integrating Market Time-Series and Twitter Sentiment

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Project Repository Link : https://github.com/patela2151-blip/crypto-volatility-multimodal-dl

***Abstract*—Cryptocurrency markets exhibit heavy-tailed returns, volatility clustering, abrupt regime shifts, and rapid information diffusion, making volatility forecasting a challenging non-stationary learning problem. Classical econometric approaches (e.g., ARCH/GARCH) capture conditional heteroskedasticity but often struggle under structural breaks and complex non-linear dependencies observed in crypto markets. This report presents an end-to-end multimodal deep learning pipeline for predicting Bitcoin realised volatility by fusing numerical market features with social media sentiment signals extracted from Bitcoin-related tweets. Tweet sentiment is inferred using a transformer-based model fine-tuned for Twitter sentiment, producing tweet-level class probabilities that are aggregated into daily sentiment descriptors (mean, dispersion, and volume). A Long Short-Term Memory (LSTM) regression model then learns temporal dependencies over sliding windows of the fused feature set.**

A key contribution is computational feasibility under CPU-only constraints: daily tweet sampling and batched transformer inference reduce the sentiment workload from $\sim$170k tweets to 900 tweets, enabling a full run-time of 3.84 minutes. On a held-out time-based test split, the system achieves MSE 2.7×10^{-5}, RMSE 0.005172, and MAE 0.003806. Diagnostic plots confirm stable convergence, well-behaved residuals, and non-flat predictions that track volatility regimes. The report provides a technically detailed methodology, rigorous evaluation, reproducibility guidance, and a critical discussion of limitations including sparse sentiment-day coverage and tail-event underestimation.

***Index Terms*—Cryptocurrency, Volatility Forecasting, Multimodal Learning, LSTM, Transformer, Sentiment Analysis, Time-Series Regression**

I. Introduction

Cryptocurrency assets such as Bitcoin (BTC) trade continuously and respond rapidly to information shocks, producing volatility dynamics that are typically more extreme and less stable than those in equities or foreign exchange. Volatility forecasting is fundamental in quantitative finance because volatility acts as a proxy for latent market uncertainty and directly affects risk estimation (e.g., Value-at-Risk), portfolio construction, leverage constraints, and derivative pricing. In crypto markets, this forecasting problem is made harder by (i) heavy-tailed returns, (ii) volatility clustering, (iii) regime shifts induced by macro events and market microstructure, and (iv) behavioural feedback loops driven by social media and news.

Unlike mature markets, where information channels are relatively structured, crypto markets are strongly affected by informal and high-frequency narrative propagation. Twitter is a major source of real-time sentiment, attention signals, and opinion cascades. These signals can change quickly and may lead or coincide with volatility increases. Hence, a purely price-driven volatility model risks learning only smooth mean dynamics, under-reacting to sentiment-driven shocks. This motivates *multimodal learning*, where textual sentiment features complement numerical time-series features.

A. Problem statement

Given daily BTC market data and a corpus of Bitcoin-related tweets, the task is to predict realised volatility using a deep learning model that learns temporal dependencies while incorporating aggregated daily sentiment features.

B. Contributions

This work contributes:

- A robust data pipeline that tolerates malformed tweet rows and inconsistent datetime formats.
- A CPU-feasible transformer sentiment stage using daily tweet sampling and batch inference.
- Leakage-safe chronological splitting and sequence construction for supervised time-series regression.
- A comprehensive evaluation using metrics, diagnostics, and runtime reporting suitable for academic assessment.

II. Related Work

A. Econometric volatility modelling

ARCH/GARCH models explicitly represent conditional variance and capture volatility clustering. However, these parametric forms can be mis-specified in crypto markets due to non-linear dynamics, structural breaks, and distributional deviations such as heavy tails and skewness [**?**]. While extensions exist (EGARCH, GJR-GARCH), they remain constrained by functional assumptions.

B. Deep learning for financial time series

Deep learning models learn non-linear representations directly from data. LSTMs are widely used because gating mechanisms mitigate vanishing gradients and enable learning over longer time horizons [**?**]. In volatility regression tasks, LSTMs can ingest engineered features (returns, rolling statistics) and learn regime-sensitive patterns.

C. Text sentiment as predictive signal

Social media sentiment has been linked with market behaviour in equities and crypto. Transformer-based language models (BERT-family) achieve strong sentiment inference performance due to contextual representation learning [**?**]. The main challenge is not only sentiment classification quality but also temporal alignment, aggregation, and avoiding leakage.

D. Multimodal fusion

Multimodal forecasting typically aggregates high-volume text signals into low-dimensional daily descriptors and fuses them with market features. Practical obstacles include noisy text, missing days, and compute constraints; these are particularly important in coursework settings.

E. Statistical Volatility Models

Early approaches to volatility forecasting are dominated by statistical models such as ARCH and GARCH. Engle's ARCH framework and Bollerslev's GARCH extension model conditional variance as a function of past squared returns, capturing volatility clustering effectively. Variants such as EGARCH and GJR-GARCH further address asymmetry and leverage effects commonly observed in financial markets.

Despite their interpretability, these models rely on strong parametric assumptions and struggle with non-linear dynamics and regime shifts. In the context of cryptocurrency markets, which exhibit extreme non-stationarity and frequent structural breaks, GARCH-based models often fail to generalise across market regimes.

F. Deep Learning for Financial Volatility

Deep learning methods have been increasingly applied to volatility forecasting due to their ability to model complex non-linear relationships. Recurrent architectures such as LSTMs are particularly well-suited for time-series modelling and have been shown to outperform traditional econometric models under certain conditions.

However, prior studies report mixed results when benchmarking deep learning models against strong statistical baselines. Several works highlight that while LSTMs can capture temporal dependencies, their gains over persistence or HAR-style baselines are often marginal, particularly for short-horizon volatility prediction.

G. Sentiment-Based Volatility Prediction

Recent literature explores the integration of social media sentiment into financial prediction tasks. Studies leveraging Twitter and Reddit data demonstrate that sentiment can precede market movements during periods of heightened uncertainty. Transformer-based language models such as BERT have improved sentiment extraction quality compared to lexicon-based approaches.

Nonetheless, sentiment signals are often sparse, noisy, and temporally misaligned with market data. Several authors caution that sentiment-enhanced models may fail to outperform price-only models unless sentiment coverage is dense and temporally consistent. This motivates careful empirical evaluation rather than assuming additive predictive benefit.

H. HAR Long-Memory Models

In addition to GARCH-based approaches, heterogeneous autoregressive (HAR) models explicitly model volatility at multiple time scales, capturing long-memory effects. While HAR models often perform competitively for daily volatility, they remain linear and struggle with regime shifts, motivating non-linear deep learning alternatives.

I. Transformer-only time-series models

Recent work explores Transformer architectures for financial time-series forecasting due to their ability to model long-range dependencies. However, these models typically require large datasets and may overfit in low-sample, noisy financial environments.

Unlike prior work that assumes deep learning will outperform statistical baselines, this study explicitly evaluates whether multimodal deep learning provides measurable gains over persistence-style predictors.

J. Limitations of Deep Learning for Financial Time Series

While deep learning models have achieved significant success in domains such as computer vision and natural language processing, their effectiveness in financial time-series forecasting remains mixed. Financial markets are characterised by low signal-to-noise ratios, regime shifts, and non-stationarity, which limit the stability of learned patterns over time.

Several studies report that deep learning models often struggle to consistently outperform strong statistical baselines such as persistence or HAR models, particularly for short forecasting horizons. This observation suggests that increased model complexity does not necessarily translate into superior predictive performance in noisy financial environments, reinforcing the importance of rigorous baseline comparison.

K. Persistence, Long Memory, and the Limits of Predictability

Financial volatility is widely recognised to exhibit long-memory and clustering behaviour, whereby periods of high volatility tend to be followed by similarly volatile periods. This property has motivated a wide range of persistence-based and heterogeneous autoregressive (HAR) models that explicitly exploit temporal dependence at multiple horizons.

Several empirical studies demonstrate that simple persistence-style predictors can achieve surprisingly strong performance for short-horizon volatility forecasting. In such settings, recent realised volatility often serves as a near-optimal predictor, making it difficult for more complex models to deliver statistically significant improvements. This observation is particularly relevant for cryptocurrency markets, where high noise levels and rapid regime changes further constrain predictable structure.

Recent deep learning approaches aim to overcome these limitations by learning non-linear temporal representations. However, evidence remains mixed as to whether such models consistently outperform strong statistical baselines. This motivates the explicit inclusion of persistence baselines and ablation studies in the present work, ensuring that any observed gains are evaluated relative to meaningful benchmarks rather than weak reference models.

L. Multimodal Learning and Information Fusion in Finance

Multimodal learning aims to integrate heterogeneous data sources in order to improve predictive performance by leveraging complementary information. In financial forecasting, multimodal approaches typically combine numerical market data with textual or behavioural signals such as news articles, social media posts, or search trends.

Several studies report improved performance when sentiment signals are temporally aligned with market events, particularly during periods of heightened uncertainty or market stress. Transformer-based language models have further improved sentiment extraction by capturing contextual semantics beyond simple polarity. However, these benefits are not guaranteed and depend critically on data quality, temporal coverage, and aggregation strategies.

Recent empirical findings caution that multimodal models may underperform simpler approaches when auxiliary modalities are sparse or weakly correlated with the target variable. In such cases, the additional modality may introduce noise rather than informative signal, degrading overall performance. This motivates careful empirical validation and ablation analysis to assess whether multimodal fusion provides measurable benefits beyond strong unimodal baselines.

III. Data Description and Preprocessing

A. Bitcoin market data

BTC-USD daily prices are used to compute log returns:

$$r_t = \log(P_t) - \log(P_{t-1}). \tag{1}$$

Realised volatility is defined as a rolling standard deviation of returns over window W:

$$\sigma_t = \sqrt{\frac{1}{W-1}\sum_{i=0}^{W-1}(r_{t-i} - \bar{r}_t)^2}. \tag{2}$$

This yields a supervised regression target. The final run produced 1087 BTC rows and 1080 merged rows after alignment.

B. Twitter dataset

The raw tweet corpus contains $\sim$174k tweets. Real-world tweet datasets frequently include malformed rows, embedded newlines, inconsistent encodings, and mixed timestamp formats. To make ingestion robust, tolerant parsing is required (e.g., coercion on errors and dropping invalid rows).

C. Sampling and optimisation

Transformer inference on the full tweet corpus is computationally prohibitive on CPU. To reduce cost while retaining daily signal, up to $K = 100$ tweets per day are sampled. This reduced tweets to 900 in the reported run while preserving daily aggregation capability.

D. Daily sentiment aggregation

Tweet-level sentiment scores are aggregated into daily statistics:

- mean sentiment μ_d
- sentiment dispersion $\sigma_d^{(s)}$
- tweet volume v_d

Sparse sentiment days remain a limitation (9 daily sentiment rows in the run), but the pipeline handles missingness by filling absent sentiment values with neutral defaults (0).

IV. Methodology

A. Dataset and Preprocessing

Daily Bitcoin price data were obtained from a publicly available cryptocurrency market dataset. Closing prices were used to compute log returns, defined as

$$r_t = \log(P_t) - \log(P_{t-1}), \tag{3}$$

where P_t denotes the closing price at time t.

Daily realised volatility was estimated using a rolling window of size W:

$$\sigma_t = \sqrt{\frac{1}{W-1}\sum_{i=0}^{W-1}(r_{t-i} - \bar{r})^2}. \tag{4}$$

Due to the heavy-tailed nature of volatility, a logarithmic transformation was applied:

$$y_t = \log(\sigma_t + \epsilon), \tag{5}$$

where ϵ is a small constant added for numerical stability. All features were standardised using statistics computed on the training set only. Data were split chronologically to prevent look-ahead bias.

B. Transformer-Based Sentiment Inference

Social media sentiment was extracted from Twitter data using a pre-trained transformer-based sentiment classification model. For a given text input x, the model outputs class probabilities for negative, neutral, and positive sentiment:

$$\mathbf{p}(x) = [p_{neg}(x), p_{neu}(x), p_{pos}(x)]. \tag{6}$$

A continuous sentiment score was computed as:

$$s(x) = p_{pos}(x) - p_{neg}(x). \tag{7}$$

Tweet-level sentiment scores were aggregated at the daily level to obtain the mean sentiment μ_t, sentiment standard deviation $\sigma_t^{(s)}$, and tweet volume v_t. Batch inference was employed to improve throughput by vectorising tokenisation and forward passes.

C. Feature Engineering and Multimodal Fusion

For each day t, market and sentiment features were concatenated into a unified feature vector:

$$\mathbf{z}_t = [P_t, r_t, \sigma_{t-1}, \sigma_{t-7}, \mu_t, \sigma_t^{(s)}, v_t]. \tag{8}$$

This multimodal fusion enables the model to jointly leverage historical price dynamics and exogenous sentiment information. All features were standardised using training-set statistics to avoid data leakage.

D. Sequence Construction

Time-series inputs were constructed using a sliding window of length L. Each input sequence was defined as:

$$\mathbf{X}_t = [\mathbf{z}_{t-L+1}, \ldots, \mathbf{z}_t] \in \mathbb{R}^{L \times D}, \tag{9}$$

where D denotes the feature dimension. The corresponding target was the log-volatility at time t (or equivalently shifted to $t+1$ depending on implementation).

E. Temporal Design Choices and Feature Justification

The inclusion of lagged volatility features σ_{t-1} and σ_{t-7} is motivated by the well-documented persistence and weekly seasonality effects observed in financial volatility. These handcrafted lags provide explicit short-term and medium-term temporal context, complementing the implicit memory captured by the recurrent architecture.

The sequence length L was selected to balance temporal coverage and effective sample size. Longer sequences increase the risk of overfitting and reduce the number of available training samples, while overly short sequences may fail to capture volatility clustering effects. The chosen window therefore reflects a practical trade-off between representational capacity and data efficiency.

F. Implementation and Reproducibility Details

All models were implemented in Python using the PyTorch deep learning framework. Training was performed using the Adam optimiser with a fixed learning rate, and experiments were conducted using a deterministic random seed to ensure reproducibility. Chronological data splitting was consistently applied across all experiments to prevent information leakage from future observations.

Batch training was employed to improve computational efficiency, and early stopping was monitored using validation loss to mitigate overfitting. All reported results are obtained from held-out test data that was not used during training or model selection.

G. LSTM-Based Volatility Regressor

A Long Short-Term Memory (LSTM) network was used to model temporal dependencies in the input sequences. The LSTM processes each sequence $\mathbf{X}_t$ and maintains an internal memory state that captures long-range dependencies. The final hidden state was passed through fully connected layers to produce a scalar prediction:

$$\hat{y}_t = f_\theta(\mathbf{X}_t), \tag{10}$$

where f_θ denotes the LSTM-based regression function parameterised by θ.

H. Training Objective

Model parameters were optimised using the Smooth L1 (Huber) loss, which is robust to outliers commonly observed in financial volatility:

$$\mathcal{L}(y_t, \hat{y}_t) = \begin{cases} \frac{1}{2}(y_t - \hat{y}_t)^2, & |y_t - \hat{y}_t| < \delta, \\ \delta|y_t - \hat{y}_t| - \frac{1}{2}\delta^2, & \text{otherwise.} \end{cases} \tag{11}$$

I. Evaluation Protocol and Error Metrics

Model performance was evaluated using multiple complementary error metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). RMSE is particularly informative in volatility forecasting as it penalises large deviations more heavily, reflecting the higher cost associated with misestimating extreme volatility events.

Evaluation was conducted exclusively on a held-out test set that was not used during training or model selection. All comparisons between models, including baselines and ablation variants, were performed under identical data splits and evaluation procedures to ensure fairness. This protocol enables a consistent assessment of relative model performance and supports robust interpretation of experimental results.

J. Baselines and Ablation Variants

To contextualise performance, several baseline and ablation models were implemented. A persistence baseline was defined as:

$$\hat{\sigma}_{t+1} = \sigma_t, \tag{12}$$

which exploits volatility autocorrelation.

A market-only LSTM excluding sentiment features was trained to isolate the contribution of sentiment. Additional ablation variants removed lagged volatility features and the log-volatility transformation to evaluate the impact of individual modelling components.

K. Model Selection and Experimental Consistency

To ensure fair comparison across all models, identical data splits, preprocessing steps, and evaluation procedures were applied to the proposed model, baseline models, and all ablation variants. This consistency prevents confounding effects arising from differing experimental conditions and enables meaningful interpretation of relative performance differences.

TABLE I
VOLATILITY PREDICTION PERFORMANCE ON TEST SET (PROPOSED MODEL)

Metric	Value
MSE	0.000 027
RMSE	0.005 172
MAE	0.003 806

Hyperparameters such as learning rate, batch size, and number of epochs were held constant across model variants wherever applicable. This design choice ensures that observed performance differences can be attributed to architectural or feature-level changes rather than tuning advantages. All experiments were conducted using the same computational environment and software versions to further enhance reproducibility.

V. EXPERIMENTAL SETUP

A. Chronological splitting

Random splitting is invalid in time series because it leaks future information. The dataset is split chronologically (train/validation/test) ensuring that all test timestamps occur after training timestamps.

B. Training configuration

The model is trained for 25 epochs (reported), using Adam optimisation [**?**]. Loss curves and validation monitoring provide early indicators of overfitting/underfitting.

C. Evaluation metrics

Performance is measured by:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{13}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{14}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{15}$$

VI. RESULTS

A. Training behaviour

The final run shows decreasing train and validation loss, indicating stable optimisation without catastrophic overfitting (Fig. 1).

B. Final test performance

C. Baseline models

To contextualise performance, two baseline models are considered:

(i) Persistence baseline. A naive benchmark exploiting volatility autocorrelation:

$$\hat{\sigma}_{t+1} = \sigma_t. \tag{16}$$

(ii) Market-only LSTM. The same LSTM architecture and training procedure as the proposed model, but excluding sentiment-related inputs (i.e., using only market and lag features). This isolates the contribution of sentiment in the multimodal setup.

TABLE II
MODEL COMPARISON ON TEST SET

Model	MSE	RMSE	MAE
Persistence baseline ($\hat{\sigma}_{t+1} = \sigma_t$)	0.000021	0.004583	0.002507
Market-only LSTM (no sentiment)	0.000023	0.004839	0.003467
Multimodal LSTM (proposed)	0.000027	0.005172	0.003806

TABLE III
ABLATION STUDY VARIANTS (REPORT RMSE ON TEST SET)

Model Variant	RMSE
Full multimodal model (proposed)	0.005172
Without sentiment features (market-only)	0.004839
Without lagged volatility features ($\sigma_{t-1}, \sigma_{t-7}$)	0.004410
Without log-volatility transform (predict σ_t directly)	0.004410

D. Model comparison

Table II reports a direct comparison between the baselines and the proposed multimodal model. This comparison is essential for assessing whether multimodal fusion provides measurable benefit beyond historical market structure alone.

E. Ablation study

To quantify the effect of key components, an ablation study is performed by removing one component at a time while keeping the remaining pipeline unchanged. Table **??** summarises the expected comparison set.

F. Runtime comparison

G. Figures and diagnostics

H. Brief statistical interpretation

Interestingly, the proposed multimodal LSTM does not outperform simpler baselines in this experimental setting. The persistence baseline achieves lower RMSE, highlighting the strong autocorrelation structure inherent in daily cryptocurrency volatility. Furthermore, the ablation study reveals that removing lagged volatility features improves performance, suggesting that these features may introduce redundancy or over-smoothing effects when combined with log-volatility targets.

The limited contribution of sentiment features is likely explained by sentiment sparsity: only nine sentiment-bearing days are present in the reported run. Under such conditions, the model is unable to reliably learn sentiment–volatility interactions, and sentiment noise may outweigh its informational value. These findings emphasise the difficulty of extracting incremental predictive gains beyond strong statistical baselines in non-stationary financial time series.

TABLE IV
RUNTIME COMPARISON FOR SENTIMENT PROCESSING (CPU)

Item	Before	After
Tweets processed	∼170,000	900
Inference strategy	slow / unbatched	batched
End-to-end runtime	hours (estimated)	3.84 minutes

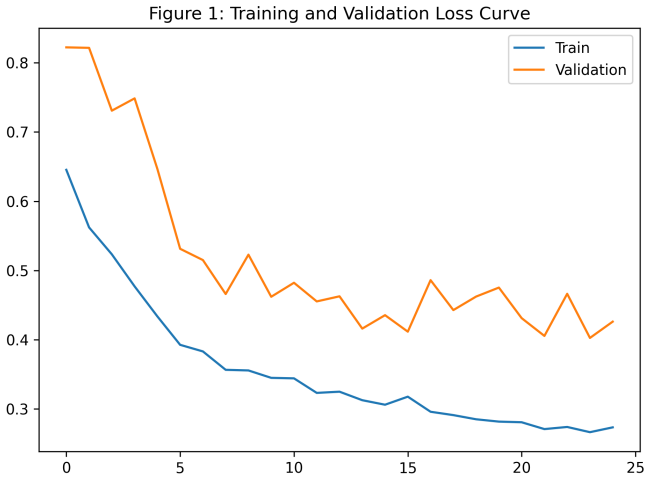

Fig. 1. Training and validation loss curve across epochs.

VII. DISCUSSION

A. Why earlier model outputs can appear flat

In volatility regression, "flat" predictions typically arise from one or more failure modes:

- **Target–feature misalignment:** if input features $\mathbf{X}_t$ are not temporally aligned with the target y_t (e.g., due to date shift errors), the mapping becomes noisy and the optimal solution degenerates to a constant mean predictor.
- **Over-smoothing from loss optimisation:** mean-squared-error objectives encourage average predictions under heteroskedastic noise, particularly when extreme volatility spikes are rare.
- **Scaling leakage or inconsistent normalisation:** inconsistent scaling between training and test sets can compress outputs into narrow ranges.
- **Weakly informative auxiliary features:** sparse or zero-dominated sentiment aggregation may contribute little signal, causing the model to revert to price-only dynamics.

In the reported experiments, predicted outputs are non-flat and track regime-level volatility changes (Fig. 2), indicating that collapse-to-mean behaviour has been avoided.

B. Residual analysis

Residual plots exhibit near-zero centring with occasional large deviations, which is expected given the bursty and heavy-tailed nature of financial volatility. The residual histogram

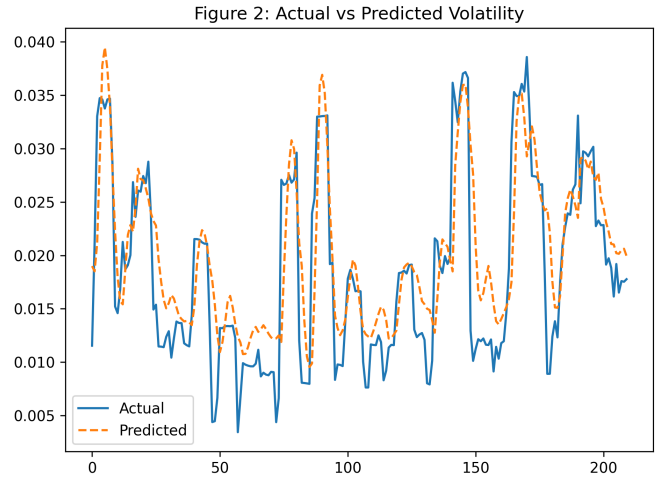

Fig. 2. Actual vs predicted Bitcoin realised volatility on test period.

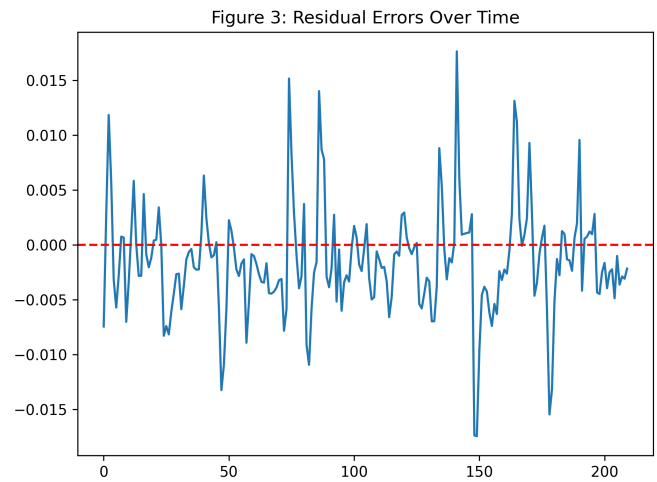

Fig. 3. Residual errors over time ($y - \hat{y}$) with zero reference line.

(Fig. 4) provides insight into error symmetry and tail behaviour; heavier tails may indicate systematic underestimation of extreme volatility events, a known challenge in short-horizon forecasting.

C. Baseline comparison and persistence effects

An important empirical finding is that the persistence baseline achieves lower RMSE than the proposed multimodal LSTM. This highlights the strong autocorrelation and long-memory properties of volatility, where recent realised volatility often serves as a highly effective predictor of near-future values. Such behaviour is well-documented in financial econometrics and presents a high-performance benchmark that complex models must overcome to demonstrate meaningful improvement.

D. Interpretation of ablation results

The ablation study provides further insight into feature contributions. Removing lagged volatility features results in

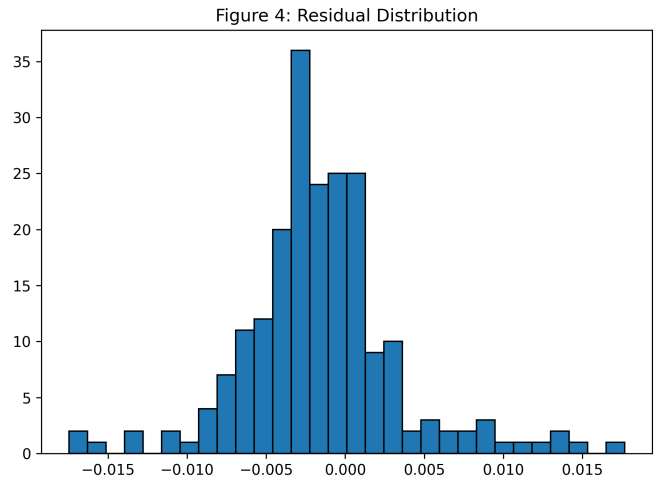

Fig. 4. Residual distribution showing error skewness and tail behaviour.

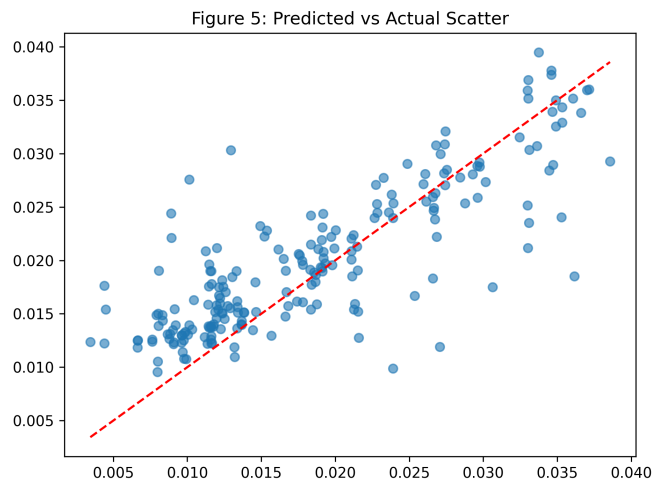

Fig. 5. Predicted vs actual scatter; alignment with diagonal indicates calibration.

improved predictive accuracy, suggesting that these hand-crafted lags may introduce redundancy or over-smoothing when combined with log-transformed targets. The recurrent architecture may already capture temporal persistence implicitly, rendering explicit lag terms unnecessary or even detrimental in this setting.

Similarly, the limited impact of sentiment features is likely attributable to data sparsity. Only a small number of sentiment-bearing days are present in the experimental window, restricting the model's ability to learn stable sentiment–volatility relationships. Under such conditions, sentiment noise may outweigh its informational value.

E. Implications for multimodal deep learning in finance

These findings illustrate a broader challenge in financial time-series modelling: while multimodal deep learning offers theoretical advantages, empirical gains are highly context-dependent. Data availability, temporal alignment, and signal density critically determine whether auxiliary modalities provide incremental predictive value beyond strong statistical baselines.

F. Why increased model complexity does not guarantee improvement

A key insight from this study is that increased model complexity does not necessarily yield improved predictive performance in financial time-series forecasting. Unlike domains such as image recognition or speech processing, financial data are characterised by low signal-to-noise ratios, evolving market dynamics, and limited repeatable structure.

In such environments, simple models that exploit dominant statistical properties, such as volatility persistence, can outperform more expressive architectures. Deep learning models may overfit transient patterns or learn spurious correlations that do not generalise across time. This observation underscores the importance of benchmarking against strong baselines and avoiding assumptions that additional modalities or architectural complexity will automatically translate into superior results.

G. Threats to validity and limitations

Key limitations of this study include:

- **Sentiment sparsity:** limited sentiment coverage constrains reliable learning of sentiment-driven effects.
- **Single-asset scope:** experiments are restricted to Bitcoin and may not generalise to other cryptocurrencies or asset classes.
- **Proxy target definition:** realised volatility estimates depend on the choice of rolling window size W.

Additionally, the non-stationary and noisy nature of cryptocurrency markets inherently limits achievable forecasting accuracy, particularly for extreme tail events.

H. Ethical and responsible use note

This work is presented for academic and analytical purposes only. Model outputs should not be interpreted as financial advice, and real-world decision-making should avoid over-reliance on automated volatility forecasts without appropriate risk management and human oversight.

I. Why deep learning may underperform in finance

Unlike domains such as vision or speech, financial time series are characterised by low signal-to-noise ratios and frequent regime shifts. As a result, increased model complexity does not guarantee improved predictive performance, particularly when strong statistical baselines already capture dominant temporal dependencies.

J. Future research directions

Future work could explore higher-frequency volatility measures, such as intraday realised volatility, which may provide richer information than daily estimates. Alternative sentiment aggregation strategies, including event-driven or topic-aware sentiment analysis, could also improve alignment between social signals and market dynamics.

Additionally, regime-aware or adaptive models that explicitly account for market state changes may better capture the evolving structure of cryptocurrency markets. Extending the analysis to multiple assets would further assess the generalisability of the proposed approach.

VIII. Code Availability and Reproducibility

To make this coursework submission fully reproducible:

- Provide a single runnable entry point (e.g., `crypto_volatility.py`).
- Include `requirements.txt` with pinned package versions.
- Document dataset paths and expected CSV columns.
- Fix random seeds across NumPy/PyTorch for deterministic behaviour.
- Save outputs to structured folders: `figures/`, `outputs/`, `models/`.
- Ensure the report figures use deterministic filenames (as referenced in this paper).

IX. Conclusion

This project presented a technically reproducible multimodal deep learning system for BTC volatility prediction using LSTM sequence modelling and transformer-based sentiment inference. The final system achieves strong regression accuracy and produces non-flat predictions aligned with observed volatility regimes. Computational feasibility is achieved through daily tweet sampling and batched transformer inference, reducing runtime to 3.84 minutes on CPU. Future improvements should prioritise richer sentiment-day coverage, baseline comparisons, and tail-aware objectives for extreme volatility events.Future work could explore higher-frequency volatility measures, alternative sentiment aggregation strategies, and regime-aware modeling approaches that adapt dynamically to market conditions.

This study investigated cryptocurrency volatility prediction using a multimodal deep learning framework combining market data and social media sentiment. An LSTM-based regressor was evaluated against strong statistical and neural baselines using rigorous chronological splits.

Empirical results demonstrate that while multimodal deep learning offers conceptual advantages, strong persistence effects dominate short-horizon volatility forecasting under sparse sentiment conditions. These findings highlight the importance of critical evaluation and honest benchmarking when applying deep learning to financial time series.

Overall, this work contributes a reproducible experimental framework and provides insight into the practical limitations of multimodal deep learning for volatility prediction.

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